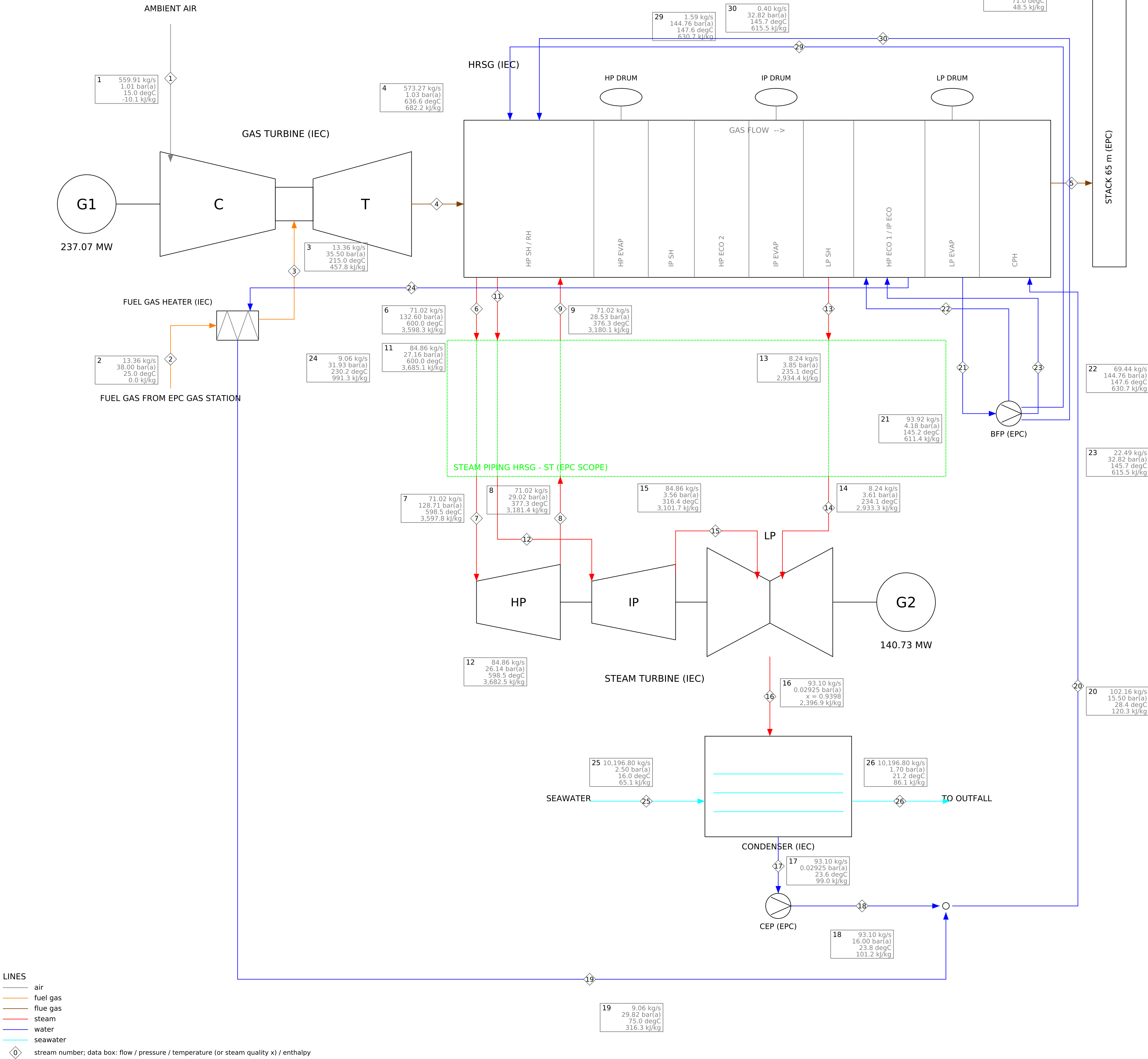




STREAM TABLE - SRC-NG-060

| No | Stream                                                              | kg/s      | bar(a) | degC / x | kJ/kg   | MW*   |
|----|---------------------------------------------------------------------|-----------|--------|----------|---------|-------|
| 1  | Ambient air to GT compressor inlet                                  | 559.91    | 1.01   | 15.0     | -10.1   | -5.7  |
| 2  | Fuel gas at performance gas heater inlet (IEC terminal point IF-01) | 13.36     | 38.00  | 25.0     | 0.0     | 0.0   |
| 3  | Heated fuel gas to GT combustion system                             | 13.36     | 35.50  | 215.0    | 457.8   | 6.1   |
| 4  | GT exhaust gas to HRSG inlet                                        | 573.27    | 1.03   | 636.6    | 682.2   | 391.1 |
| 5  | Flue gas at HRSG outlet to stack                                    | 573.27    | 1.01   | 71.0     | 48.5    | 27.8  |
| 6  | HP steam at HRSG outlet (HP SH outlet)                              | 71.02     | 132.60 | 600.0    | 3,598.3 | 255.6 |
| 7  | HP steam at ST main stop valve inlet                                | 71.02     | 128.71 | 598.5    | 3,597.8 | 255.5 |
| 8  | Cold reheat steam at HP turbine exhaust                             | 71.02     | 29.02  | 377.3    | 3,181.4 | 226.0 |
| 9  | Cold reheat steam at HRSG reheater inlet                            | 71.02     | 28.53  | 376.3    | 3,180.1 | 225.9 |
| 10 | IP superheated steam (joins cold reheat inside the HRSG)            | 13.43     | 29.19  | 310.9    | 3,024.1 | 40.6  |
| 11 | Hot reheat steam at HRSG outlet                                     | 84.86     | 27.16  | 600.0    | 3,685.1 | 312.7 |
| 12 | Hot reheat steam at ST IP stop/control valve inlet                  | 84.86     | 26.14  | 598.5    | 3,682.5 | 312.5 |
| 13 | LP steam at HRSG outlet                                             | 8.24      | 3.85   | 235.1    | 2,934.4 | 24.2  |
| 14 | LP steam at ST LP admission                                         | 8.24      | 3.61   | 234.1    | 2,933.3 | 24.2  |
| 15 | IP turbine exhaust (crossover, internal)                            | 84.86     | 3.56   | 316.4    | 3,101.7 | 263.2 |
| 16 | LP turbine exhaust to condenser (UEEP)                              | 93.10     | 0.0293 | x 0.940  | 2,396.9 | 223.1 |
| 17 | Condensate at hotwell outlet / CEP suction                          | 93.10     | 0.0293 | 23.6     | 99.0    | 9.2   |
| 18 | Condensate at CEP discharge                                         | 93.10     | 16.00  | 23.8     | 101.2   | 9.4   |
| 19 | Fuel gas heater water return to condensate                          | 9.06      | 29.82  | 75.0     | 316.3   | 2.9   |
| 20 | Condensate to HRSG condensate preheater inlet                       | 102.16    | 15.50  | 28.4     | 120.3   | 12.3  |
| 21 | Feedwater from LP drum to boiler feed pumps                         | 93.92     | 4.18   | 145.2    | 611.4   | 57.4  |
| 22 | HP feedwater at BFP discharge to HP economiser                      | 69.44     | 144.76 | 147.6    | 630.7   | 43.8  |
| 23 | IP feedwater at BFP interstage to IP economiser                     | 22.49     | 32.82  | 145.7    | 615.5   | 13.8  |
| 24 | IP economiser water to performance fuel gas heater                  | 9.06      | 31.93  | 230.2    | 991.3   | 9.0   |
| 25 | Circulating seawater to condenser inlet                             | 10,196.80 | 2.50   | 16.0     | 65.1    | 663.9 |
| 26 | Circulating seawater from condenser outlet                          | 10,196.80 | 1.70   | 21.2     | 86.1    | 878.0 |
| 29 | HP attenuation spray water (BFP discharge)                          | 1.59      | 144.76 | 147.6    | 630.7   | 1.0   |
| 30 | Reheat attenuation spray water (BFP interstage)                     | 0.40      | 32.82  | 145.7    | 615.5   | 0.2   |

\* energy flow m x h (reference: IAPWS-IF97 for water/steam, 25 degC sensible for gases, cp x T for seawater)

PLANT POWER BALANCE

|                                       |        |        |
|---------------------------------------|--------|--------|
| GT output (generator terminals)       | 237.07 | MW     |
| ST output (generator terminals)       | 140.73 | MW     |
| Gross output                          | 377.80 | MW     |
| Auxiliary loads                       | -6.64  | MW     |
| Transformer losses (GSU, UAT, MV/LV)  | -1.06  | MW     |
| NET OUTPUT (HV terminals)             | 368.10 | MW     |
| Heat input (LHV)                      | 634.14 | MW     |
| Gross heat rate (LHV)                 | 6,043  | kJ/kWh |
| NET HEAT RATE (LHV)                   | 6,202  | kJ/kWh |
| Net efficiency (LHV)                  | 58.05  | %      |
| Fuel LHV (ISO 6976, from composition) | 47.48  | MJ/kg  |
| Fuel gas flow                         | 65,639 | Sm3/h  |
| Circulating water flow                | 36,112 | m3/h   |
| GT load                               | 55.78  | %      |
| PLANT LOAD (NET, % OF SRC-NG-100)     | 60.00  | %      |

CHECKS (guarantees, limits, balances)

|    |                                                                                            |
|----|--------------------------------------------------------------------------------------------|
| OK | energy balance CMIX: residual -0.004 MW (tolerance 0.25 MW)                                |
| OK | energy balance COND: residual -0.184 MW (tolerance 0.89 MW)                                |
| OK | energy balance FGH: residual +0.002 MW (tolerance 0.25 MW)                                 |
| OK | energy balance GT: residual +0.013 MW (tolerance 0.63 MW)                                  |
| OK | energy balance HRSG: residual +1.482 MW vs stated casing loss 1.453 MW (tolerance 0.69 MW) |
| OK | energy balance PIPE: residual +0.362 MW (tolerance 0.82 MW)                                |
| OK | energy balance ST: residual -0.002 MW (tolerance 0.59 MW)                                  |
| OK | gross heat rate 6,042.65 kJ/kWh vs IG-07 <= 6,060.00 (margin +0.29 %)                      |
| OK | net heat rate 6,201.82 kJ/kWh vs PG-03 <= 6,250.00 (margin +0.77 %)                        |

|          |                                                                               |
|----------|-------------------------------------------------------------------------------|
| PROJECT  | Alpilla 600 MW Combined Cycle Power Plant                                     |
| TITLE    | HEAT AND MASS BALANCE - SRC-NG-060                                            |
| CASE     | Site Reference Conditions, natural gas, 60 % plant net load (guarantee PG-03) |
| DOCUMENT | ALP-HMB-SRC-NG-060                                                            |
| STATUS   | preliminary - generated by engine hmb v1.1.0                                  |
| DATA     | a363d729afc29597   2026-09-28                                                 |
| NOTE     | all values calculated from the database; do not edit                          |